

Vector Space

→ A vector space is a collection of objects called vectors, along with two operations: vector addition and scalar multiplication, which satisfy a specific set of rules or axioms.

Formally, A vector space V over a field F

(i) For any two vectors u and v in V , their sum $u+v$ is also in V .

(ii) For any scalar c in F and any vector v in V , the product cv is also in V .

Also, there are other axioms that is true in a vector space:

(i) $u+v = v+u$

(ii) $u+(v+w) = (u+v)+w$

(iii) $0+u = u+0 = u$ [0 is zero vector]

(iv) $u+(-u) = (-u)+u = 0$

(v) $k(u+v) = ku + kv$

(vi) $(k+m)u = ku + mu$

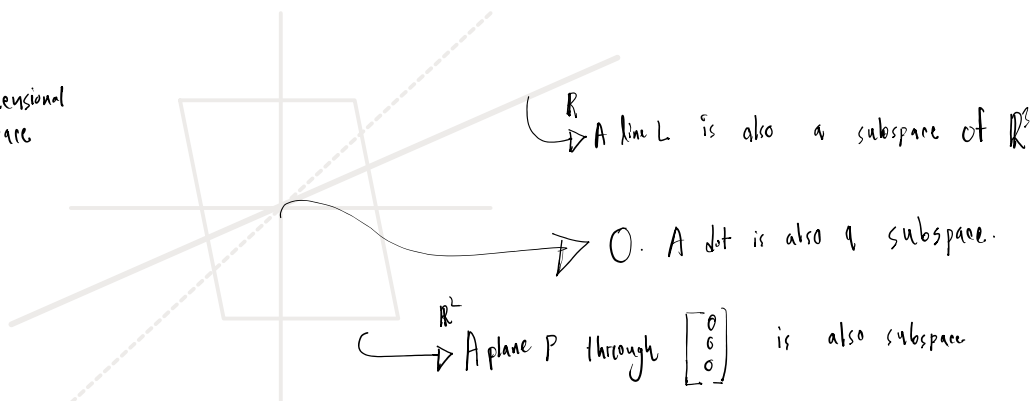
(vii) $k(mu) = (km)u$

(viii) $1u = u$.

A deeper look into vector space and their subspace

$\mathbb{R}^3$

→ three dimensional vector space



→ $\mathbb{R}^3$ is a space, inside $\mathbb{R}^3$ there will be vector spaces of their own.

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→ Question: Is the union of $\overset{(P \cup L)}{P}$ and L are a subspace of $\mathbb{R}^3$?

No This is not closed under addition.

If you take the union of the plane P and line L , you get the set of all vectors that lie either on P or on L . Assume, $u \in P$ and $v \in L$. Their sum, $u+v$, will generally not lie entirely in P or in L , but somewhere else in the space (outside both the plane and the line).

Thus, the union of P and L are not subspace.

Question: Is the intersection $(P \cap L)$ are a subspace of $\mathbb{R}^3$?

Yes The set $P \cap L$ is the set of all vectors that are in both P and L .

Assume, $u \in P \cap L$ and $v \in P \cap L$, they both are in subspaces P and L .

Then, $u+v$ must also be in both P and L . Thus, $u+v \in P \cap L$.

The same can be applied for multiplication rule.

* A subspace containing v and w must contain all linear combinations $cv+dw$

Column Space

↓ of a matrix is the set of all possible linear combinations of its column vectors. In other words, it is the span of the column vectors of the matrix.

Example 1: $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$

So, any vector b in the column space can be written as: $b = c_1 a_1 + c_2 a_2$.

This matrix signifies that a plane in $\mathbb{R}^3$ because the two columns are vectors in $\mathbb{R}^3$ and their linear combinations span a two dimensional subspace (a plane).

Example 2: $A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix}$

Signifies column space of A is subspace of $\mathbb{R}^4$

But there are one dependent column. So their linear combination a two-dimensional subspace (a plane).

* The column space consists of all linear combinations of the columns.

Those combinations are all possible vectors Ax . They fill the column space $C(A)$.

To solve $Ax=b$ is to express b has to be in the column space produced.

→ Here, the vector b must be a linear combination of the columns of A .

So, if b can be written as a combination of the columns of A , there is a solution.
else, there is no solution.

Row Space

→ of a matrix is the set of all possible linear combinations of its row vectors.
It is a subspace of $\mathbb{R}^n$ and is characterized by the following properties:

(i) The row space is spanned by the nonzero rows of the matrix.

(ii) The dimension of the row space is called the rank of the matrix, which represents the maximum number of linearly independent rows.

→ The rows of matrix can be the column space of matrix A^T .

→ Row space is a subspace of $\mathbb{R}^n$ (n = number of columns)

→ $A^T y = c$ to have a solution, the vector c must

$\hookrightarrow A^T y = c$ to have a solution, the vector c must lie within the column space of A^T . Basically c needs to be a linear combination of the columns of A^T .

Example:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$\hookrightarrow \text{REF of } A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{bmatrix}$$

To find the row spaces of a matrix:

- $\rightarrow$ Find REF
- $\rightarrow$ Find non-zero rows
- $\rightarrow$ Find the rank.

Here, the non zero rows are $(1, 2, 3)$, $(0, -3, -6)$

These rows form a basis for the row space of the original matrix A .

Since there are only two linear independent rows, the rank of A is 2.

So, the row space is a 2-dimensional subspace of $\mathbb{R}^3$.